

Human-in-the-Loop Dermatology AI:

Combining Reinforcement Learning and Explainable AI for Diagnosis, Treatment, and Prioritization

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Concept Note

Background & Problem Statement

Dermatological AI has demonstrated diagnostic capability but suffers from three critical limitations in the Ugandan context:

1. **Phenotypic bias:** –models trained on ISIC data (Fitzpatrick I–III) underperform on darker skin (Fitzpatrick IV–VI) [1, 5];
2. **Transparency gaps:** – deep learning outputs opaque probability scores without clinical reasoning [8];
3. **Specialist shortage:** – skin diseases account for over 50% of outpatient visits in Uganda while fewer than 20 dermatologists serve the entire population [2, 3].

Traditional static models cannot adapt to local disease patterns, including HIV-related Kaposi Sarcoma prevalent in the region [2].

Research Aim & Objectives

NOTE: To develop and evaluate a Human-in-the-Loop (HITL) system combining Equity-Driven Reinforcement Learning with Explainable AI (XAI) to improve fairness, transparency, and accuracy of dermatological triage, diagnosis, and treatment for Ugandan patients across all skin tones.

- O1 Explainability:** Integrate Concept Bottleneck Models (CBMs), Integrated Gradients, and counterfactuals for clinical-term explanations [9, 10].
- O2 Clinical Validation:** Deploy the HITL interface at Mulago Hospital with daily clinician override-driven retraining [7].
- O3 Technical Development:** Engineer a PPO triage agent with an equity-weighted reward achieving Fitzpatrick I–VI parity [5, 6].

Methodology

A four-stage HITL pipeline:

- Stage 1 : PPO RL agent** with an equity-driven reward, where false negatives on dark skin are penalised $\lambda = 10\times$ more heavily:

$$R(s, a) = \underbrace{\text{Acc}(s, a)}_{\text{diagnostic}} - \lambda \cdot \mathbb{I}[\text{FN} \wedge \text{Fitzpatrick IV–VI}], \quad \lambda = 10. \quad (1)$$

- Stage 2 : XAI layer** using HRNet features + Concept Bottleneck Models + saliency maps + counterfactuals [9, 10, 8].

- Stage 3 : Clinician Feedback Loop** via a real-time web dashboard (React + FastAPI + WebSocket), where *Accept/Override* actions log reward signals.

- Stage 4 : Offline Policy Refinement** running nightly (overrides $\rightarrow$ reward recompute $\rightarrow$ 50 PPO epochs $\rightarrow$ A/B gate).

Training uses ISIC 2019/2020 [4] blended with **2,400 curated Ugandan clinical photographs** in a custom Gymnasium simulator. Prototype prospective validation will be conducted at Mulago National Referral Hospital (Dermatology + HIV clinics) via a stepped-wedge RCT.

Expected Results

- Phenotypic bias gap closed to $\approx 0\%$ across Fitzpatrick types;
- **+22%** melanoma / Kaposi Sarcoma detection in resource-limited settings;
- Malignant mismanagement reduced from **58.8% to 4.1%**;
- Clinician trust $> 80\%$ acceptance;
- Calibration error (ECE) < 0.05 ;
- The static tool evolves into a continuously learning diagnostic partner.

Evaluation Metrics

Table 1: Evaluation metrics and target thresholds.

Metric	Target
Diagnostic accuracy (vs. histopathology)	$\geq 88\%$
Melanoma recall (Fitzpatrick IV–VI)	$\geq 95\%$
Kaposi Sarcoma F1	≥ 0.90
Bias gap (Δ Accuracy, Fitzpatrick I–VI)	$\leq 3\%$
Calibration error (ECE)	≤ 0.05
Malignant mismanagement rate	$\leq 5\%$
Clinician agreement (Fleiss’ κ)	≥ 0.75
Time-to-decision	≤ 45 s
Clinician override rate	15–25%

Alignment

- **SDG 3** (Health), **SDG 9** (Innovation), **SDG 10** (Equity).
- **NDP IV:** Human Capital Development, Digital Transformation, STI.
- **Vision 2040:** Transition from technology consumer to developer; force-multiplier for specialists; equitable access.

Next Steps

Table 2: Implementation roadmap.

Phase	Activity
Phase 1	Retrospective validation on 500 Mulago cases.
Phase 2	Prospective RCT pilot at 2 clinics.
Phase 3	Regulatory pathway (NDA + FDA SaMD).
Phase 4	Federated learning across 5 hospitals.
Phase 5	Patient-facing USSD/SMS self-triage.

Key References

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